

AI Usage Report

Course: _____SOFTWARE FOR MOBILE DEVICES (SMD)_____

Assignment: _____ASSIGNMENT 1_____

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Date: _____3-OCTOBER-2026_____

1. AI Tool(s) Used

List the AI tools used during the development of your application.

Example: ChatGPT, Gemini, GitHub Copilot

Response:

I used **Google Gemini** (via the Antigravity IDE assistant) as an AI coding assistant throughout the development of this assignment.

2. Purpose of AI Usage

Explain what you used AI for during the development of your application.

Examples:

- Understanding React/React Native concepts
- Debugging errors
- Generating or improving UI ideas
- Implementing charts using react-native-chart-kit
- Understanding unfamiliar code
- Improving code structure

Response:

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- **Planning:** Initial planning and modules selection.
 - **Debugging errors:** a SyntaxError: Identifier 'adviceTitle' has already been declared error in CourseCard.js caused by a let re-declaration inside the same scope.
 - Understanding why **react-native-chart-kit's BarChart component** crashes on the web target and finding a compatible alternative using react-native-svg.
 - **Refactoring code for readability** — replacing inline ternary operators and Math.max()/Math.min() calls with explicit if-else blocks so the logic is easier to explain and follow.
 - **Understanding how react-native-svg primitives work** (Rect, Line, Text) to build a custom bar chart component.

- **Understanding component prop-passing patterns** in React Native, such as how DashboardScreen passes threshold, courses, and width down to the chart components.

3. Important Prompts Used

Provide the important prompts that you gave to the AI tool.

Prompt 1:

I want to create an attendance based system implemented in react native, i want to create modules like Attendance, Marks and the user should be able to create tasks that he can check later on, like to log attendance per course, enter marks, and create tasks they can check

I want a single theme file that all my project files would follow, I want dark and sharp colors for my application. For creativity standpoint i wanna add a What-If Calculator that calculates if the student's attendance can be saved if he is attending or missing future classes,

A single dashboard page would show notifications of pending tasks or attendance below a threshold or marks below a threshold, I also dont wanna use any navigation bar so we will use simple divs and backbuttons, also i want charts of semester health, attendance bar charts and and the amount of work that is pending,

We will also display meaningful hardcoded messages based on what the students' marks or attendance values are.

I want components for everything as it can be used later on. Since no database required we shall use objects and arrays in javascript hardcoded.

I have added initial files and logic , but you need to add some styling and calculations logic. Create an implementation plan first before going forward. I have also added Expo MCP for the use.

Prompt 2:

"I'm getting SyntaxError: Identifier 'adviceTitle' has already been declared at line 91 in CourseCard.js. The variable is already declared with let earlier in the function. How do I fix this without changing the logic?"

Prompt 3:

"The BarChart from react-native-chart-kit is crashing on web with a render error. The call stack points to AttendanceBarChart.js. Can you replace it with a working custom chart using react-native-svg that keeps the same behavior — coloring bars red if below threshold, green if above?"

Prompt 4:

"Can we refactor the logic in the components/ folder so that ternary operators are replaced with if-else blocks, and Math.max()/Math.min() are replaced with equivalent if-else comparisons? Keep the logic identical so nothing breaks, just make it more readable and easier to explain."

4. AI-Generated Output

Briefly describe the code, solution, or suggestion provided by the AI.

Response:

Prompt 1 (Planning):

The AI helped design the overall component and screen structure. It suggested splitting the app into screens/ (Dashboard, Attendance, Marks, Tasks) and components/ (CourseCard, TaskCard, StatCard, ProgressBar, WhatIfCalculator, chart components, etc.), with a single theme.js exporting colors, spacing, typography, and radius tokens. It also outlined the What-If Calculator logic—calculating simulated attendance percentage if a student attends or skips a given number of future classes and proposed the threshold-based alert system on the Dashboard.

Prompt 2:

The AI explained that in JavaScript, declaring a variable with let twice in the same function scope causes a SyntaxError. It identified that adviceTitle was first declared on line 42 of CourseCard.js and then incorrectly re-declared with let further down. The fix was to remove the second let keyword, keeping the assignment but not the declaration.

Prompt 3:

The AI replaced the entire AttendanceBarChart.js component. Instead of importing BarChart from react-native-chart-kit, it built a custom chart using react-native-svg's <Svg>, <Rect>, <Line>, and <Text> primitives. It calculated bar heights proportionally from a 0–100% y-axis range and colored each bar based on whether the course's attendance was above or below the threshold. Y-axis grid lines at 0, 25, 50, 75, 100 were added, along with value labels on top of each bar.

Prompt 4:

The AI rewrote conditional expressions in components like CourseCard.js and WhatIfCalculator.js. For example, `const remaining = Math.max(0, totalPlanned - held)` became:

```
let remaining = totalPlanned - held;
if (remaining < 0) {
  remaining = 0;
```

```
}
```

And nested ternaries for determining `statusTone`, `statusLabel`, and `adviceTitle` were expanded into a readable `if / else if / else` chain.

5. Changes Made by Me

Explain what you changed, modified, or improved in the AI-generated solution.

Response:

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- I reviewed the AI-generated `AttendanceBarChart.js` and adjusted the `PADDING_LEFT`, `PADDING_BOTTOM`, and `PADDING_TOP` constants to better fit the actual screen layout in my app.
 - I verified that the `colors.danger` and `colors.safe` values from my `theme.js` were being correctly referenced and matched the rest of the app's color scheme.
 - For the refactoring task, I read through each changed component to confirm the logic was identical to before — particularly the attendance percentage thresholds and the "what-if" simulation math in `WhatIfCalculator.js`.
 - I kept all existing comments in the refactored code and added a few of my own where I felt a step needed clarification.
 - I adjusted variable names in a couple of places where the AI used generic names, replacing them with names consistent with the rest of the codebase.
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6. My Understanding

Explain the important part(s) of the AI-generated code in your **own words**.

Response:

Custom Bar Chart (`AttendanceBarChart.js`):

The chart works by first filtering out courses that have no classes held yet. For each remaining course, it calculates an attendance percentage. The SVG drawing area is divided into equal slots — one per course. Each bar's height is computed as $(\text{value} / 100) * \text{drawHeight}$, where `drawHeight` is the pixel height of the chart area. The bar color is chosen with a simple `if` check: if the value is below threshold, use `colors.danger` (red); otherwise use `colors.safe` (green). The `<Rect>` element draws the filled bar, while `<SvgText>` elements draw the percentage label above the bar and the course code label below it.

Re-declaration Error:

In JavaScript, `let` and `const` are block-scoped but cannot be re-declared in the same function scope. If a variable is declared with `let` at the top of a function, using `let` again for the same name anywhere

else in that same function — even after an if block — throws a `SyntaxError` at parse time. The fix is straightforward: only use `let` once for the initial declaration, then use plain assignment (`=`) everywhere else.

7. Verification and Testing

Explain how you tested the AI-generated solution.

Mention:

- Whether the code worked correctly
- Errors encountered
- How you fixed the errors
- Any changes required after testing

Response:

- **BarChart fix:** After replacing `react-native-chart-kit`'s `BarChart` with the custom SVG implementation, I pressed `r` in the Expo terminal to clear the Metro bundler cache and reloaded the web browser. The chart rendered correctly with colored bars, percentage labels, and a working legend.
 - **Variable re-declaration fix:** After removing the duplicate `let adviceTitle` declaration in `CourseCard.js`, the Metro bundler compiled without errors and the Attendance screen loaded correctly with all course cards displaying the right status and advice message.
 - **Refactoring verification:** I tested each screen after the refactor — Dashboard, Attendance, Marks, and Tasks — to confirm behavior was unchanged. The attendance calculations, "what-if" simulator, and threshold-based color coding all produced the same results as before.
 - **Errors encountered:** The main error encountered was that `react-native-chart-kit` is not compatible with `react-native-web` because it internally uses SVG APIs that are mapped differently in the web environment. The solution was to drop the library for this component and use raw `react-native-svg` instead, which is officially supported on web.
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8. Reflection

What did you learn from using AI during this assignment?

Response:

Using AI during this assignment helped me move past blocking errors quickly instead of spending hours reading library documentation. The most useful interaction was understanding *why* `react-native-chart-kit` was broken on web — the AI explained the underlying compatibility issue

clearly, which helped me understand the difference between how SVG is handled in native vs. web environments in a React Native project.

The refactoring task was also valuable: by asking the AI to convert ternary expressions to if-else blocks, I ended up reading the logic more carefully and genuinely understanding the attendance threshold calculations that were previously hidden inside one-liners. I now understand how the pct variable flows through CourseCard.js to determine the status pill color, the progress bar fill, and the advice message.

Overall, I treated the AI as a knowledgeable collaborator — it could suggest solutions and explain concepts, but I was responsible for understanding, verifying, and integrating the output into my codebase.

Student Declaration

I confirm that I have used AI tools only as a development assistant and that I understand the code submitted as part of this assignment. I am able to explain and demonstrate the functionality of my application.

Student Name: ___uzair majeed_____

Signature: _____Uzair_____

Date: _____3-OCTOBER-2026_____